

Venkatraman Raguraman

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PROFESSIONAL SUMMARY

Principal Product Manager (Tech) with 17+ years shipping AI-powered enterprise SaaS, MES/ERP, and supply-chain platforms across life sciences, energy, and automotive. Builder of agentic AI workspaces, LLM-powered copilots, and multi-agent orchestration for production, quality, and engineering workflows deployed to 15+ regulated factories. Track record translating complex domain workflows into clean data models, scalable APIs, and adopted products — combining deep operations, supply chain, Order-to-Cash, and Lean/Six Sigma fluency with modern AI product practice (RAG, fine-tuning, evaluations, guardrails, human-in-the-loop). Six Sigma Black Belt, CSCP.

CORE SKILLS

AI Product Management: Agentic AI, multi-agent orchestration, LLM application design, RAG, fine-tuning, prompt engineering, evaluation frameworks (LLM-as-judge), guardrails, responsible AI, human-in-the-loop, copilot UX, AI observability

Product Strategy: Product roadmap & vision, OKRs/KPIs, Go-to-Market (GTM), Agile/Scrum, sprint planning, backlog refinement, customer discovery, pilot governance, release management

Manufacturing & Supply Chain: MES, QMS, PLM, digital twin, IoT, BOM & change orders, genealogy/lot-serial tracking, CAPA, OT-IT integration, GxP & 21 CFR Part 11, demand forecasting, S&OP, Order-to-Cash, CPQ, dealer/partner portals

Data & Analytics: SQL, Power BI, Tableau, adoption dashboards, KPI design, Oracle Analytics

Continuous Improvement: Six Sigma Black Belt (DMAIC), Lean, Kaizen, value stream mapping, square root costing, Kanban, Min-Max

Leadership: Cross-functional facilitation, offshore team leadership, stakeholder engagement, change management, executive communication

EXPERIENCE

Oracle Inc. | Principal Product Manager (Tech)

Redwood City, CA · Aug 2021 – Present

Own roadmap, requirements, and delivery for Manufacturing Cloud — AI-native MES/QMS, OT-IT integrations, and embedded GenAI features used by large discrete and life-sciences manufacturers across NA, EU, and APAC.

- **Shipped AI-native workspaces and 8 autonomous agents** — Disposition Assistant, Resource Change Assistant, Work Order Change Assistant, Operations Assignment Assistant, Procedure Advisor (RAG copilot), and Authoring Assistant — using LLMs, agent orchestration, and tool-use APIs to automate change-order propagation, quality dispositions, corrective actions, BOM/routing management, and work-order assignments; **adopted across 10+ factory sites.**
- **Designed multi-agent orchestration platform for shift supervisors** with specialized agents for exception triaging, anomaly detection, operator assignment, and AI-drafted shift reports; grounded in real-time OT data, workforce qualification records, and historical exception libraries via RAG pipelines.
- **Built AI-powered genealogy and quality investigation workspace** for lot/serial tracked products — auto-generates CAPA resolutions, surfaces related non-conformances, and produces GxP-compliant audit reports with electronic signature under **21 CFR Part 11.**
- **Defined evaluation frameworks (LLM-as-judge), guardrails, and human-in-the-loop checkpoints** for agent reliability in regulated environments; led prompt engineering and fine-tuning on manufacturing domain datasets; partnered with applied science to reduce hallucinations on engineering and quality workflows.
- **Implemented digital-thread experience** with 3D work instructions, redlining, change-order flows, and genealogy tracking with electronic signature and audit trails — adopted by life-sciences and medical-device customers under GxP and 21 CFR Part 11.
- **Led OT-IT and HR-system integrations** defining integration architecture, APIs, and data-migration patterns for skill-based workforce planning, qualification tracking, and real-time process monitoring.

- **Drove 3-year MES/QMS and Digital Twin execution program** delivered to 15+ factories; managed product roadmap, release cadence, pilot governance, and customer rollouts converting to recurring SaaS subscriptions; achieved **60% cloud SaaS fleet growth**.

Cummins Inc. | Senior Product Manager (Tech)

Minneapolis, MN · Feb 2013 – Aug 2021

Led product and program initiatives across Lead-to-Cash, dealer/partner portals, CPQ, and industrial manufacturing operations for a global power-systems business. Six Sigma Black Belt driving cross-functional delivery and process transformation.

- **Launched Salesforce CPQ** for Engineer-to-Order (ETO) and Configure-to-Order (CTO) generators and switches; reduced quoting cycle time ~8% and contributed **~\$40M revenue** in bid-to-spec markets; built Power BI dashboards tracking quote-to-order conversion across 1,000+ partner organizations.
- **Delivered B2B dealer portal** enabling marketing, sales, ordering, and fulfillment across 5 NA regions; grew channel revenue from **~\$20M to ~\$90M (4.5x) in 18 months** and onboarded 1,000+ dealers.
- **Improved demand forecasting by +40% MAP** by integrating option-mix planning percentages from CRM-quote, lead systems, and consultant sizing into S&OP, enabling more accurate production planning.
- **Built product-data hub** auto-converting BOM configuration data into commercial price books — reducing translation errors by **98%** and eliminating cost from mis-quoted orders.
- **Portfolio Optimization via square root costing** with Finance — quantified fully-loaded cost of product options; delivered **~\$800K first-year savings** via option rationalization; governance framework adopted for future portfolio reviews.
- **Lean value-stream optimization (DMAIC)** across generator assembly — identified 7 waste categories, rebalanced to takt time, introduced Kanban/Min-Max stocking; lifted production efficiency **50%** and reclaimed **~20,000 sq ft** repurposed for a new product line.
- **Cross-functional quality initiatives** reduced field defects **70%** through structured root-cause analysis and corrective-action tracking.
- **RV Industry RECT Consulting — Chairman's Award:** Led cross-industry engagement with RV OEMs, dealer associations, and IT vendors; conducted dealer fieldwork; built nationwide industry dashboard driving **40–50% reduction in repair event cycle time**.

Ashok Leyland Ltd. | Deputy Manager, Supply Chain

Chennai, India · Aug 2008 – Aug 2011

- **Stood up NPI casting supply base (~\$12M)** for five plants in three months, enabling on-time ramp-up without disrupting peak production.
- **Set up automated machining and assembly lines** and trained 30 operators in a unionized environment; achieved target throughput in four months at **<100 PPM** defect rate.

EDUCATION

M.S., Industrial Engineering and Operations Research | University of Michigan | Dec 2012

B.Tech., Production Engineering | National Institute of Technology | Jun 2008

CERTIFICATIONS

Six Sigma Black Belt (SSBB) & Green Belt — Cummins | Certified Supply Chain Professional (CSCP) - APICS/ASCM | Generative AI OCI Foundations Associate — Oracle | Generative AI - DeepLearning.AI | Certified Scrum Master (CSM) & Product Owner (CSPO) - Scrum.org | ISO 9008 Auditor — Bureau Veritas